Short Paper

AIAER-Mask: Safe Neuro-Symbolic Intent-Correct Routing for Event-Driven Microservices

Hossam Hassan, Shadia Sarhan, and Ahmed Shalaby

***Abstract*—Event-driven microservice systems route heterogeneous operations through rapidly changing service replicas. Conventional performance-oriented routing systems, including static routing, EWMA-based selection, Power-of-Two-Choices, and unconstrained reinforcement-learning policies, can select low-latency replicas that are semantically invalid for the requested operation. Such routing mistakes may preserve raw throughput while violating application intent, consistency, privacy boundaries, or update semantics. This paper presents AIAER-Mask, a safe neuro-symbolic routing framework for intent-correct event routing. AIAER-Mask separates safety from adaptation: a lightweight symbolic intent guard derives a hard action mask from a capability map, while an offline discrete Soft Actor-Critic policy selects among only admissible replicas using telemetry features. The symbolic mask is not a neural model; it is a deterministic safety layer that prevents invalid actions by construction under the active validated map. The evaluation covers multi-replica routing, ablations, masked versus unmasked baselines, failure injection, dynamic/adversarial scenarios, broker-delay and local region-latency emulation, hyperparameter sensitivity, training diagnostics, reproducibility reporting, and DeathStarBench SocialNetwork/HotelReservation cross-workload evaluation. Across both workloads, hard symbolic masking preserves 100% intent correctness and 0% invalid actions for masked variants under validated and actively enforced capability maps, while performance-only no-mask baselines violate semantic routing constraints heavily. The mask itself is low-overhead for heuristic routers, whereas most additional overhead in full AIAER-Mask comes from neural inference, telemetry/state processing, and feature handling. Strong masked heuristics such as Mask+EWMA and Intent-safe P2C remain highly competitive and sometimes outperform the learned policy in latency and regret. We therefore position AIAER-Mask as a safety-constrained neuro-symbolic framework, not as the fastest router in every regime. The current evidence supports instantiation across two single-region DeathStarBench workloads and local region-like latency profiles; broader production-trace, real multi-broker, and real multi-region validation remains future work.**

***Index Terms*—Event-driven architecture, microservices, safe routing, neuro-symbolic systems, offline reinforcement learning, symbolic masking, intent correctness, DeathStarBench, reproducibility.**

I. Introduction

EVENT-DRIVEN microservice architectures decompose user-facing operations into many asynchronous service interactions. A request such as composing a social-network post or reading a timeline may traverse event brokers, edge routers, cache layers, and multiple service replicas. Routing therefore affects latency, throughput, and tail behavior. However, performance is not the only requirement. Each operation carries semantic intent: a write operation must not be routed to a read-only handler, a timeline read must not be handled by a media-only endpoint, and an operation scoped to one capability group must not be sent to a replica outside that group.

This distinction exposes a weakness in common routing practice. Static routing, EWMA latency routing, Power-of-Two-Choices, and learned policies can all optimize latency while ignoring semantic admissibility. A globally fast replica is not necessarily a valid replica for the current operation. Treating intent correctness as a soft reward penalty is also insufficient, because a low-latency invalid action can still be selected when the learned policy is uncertain, the reward is misspecified, or the deployment workload shifts. In safety-sensitive or consistency-sensitive microservice systems, the correct question is not only which service is fastest, but which services are allowed.

Existing work on cloud routing, microservice load balancing, and reinforcement learning for systems has shown that adaptive policies can improve resource use under changing load [5], [8], [9]. Offline reinforcement learning avoids unsafe online exploration by learning from historical behavior logs [3], [4]. Intent-based networking and zero-touch management emphasize high-level operator goals [15], [16]. Yet these lines of work do not directly provide a microservice replica-level mechanism that guarantees semantic admissibility for every routing action while still allowing adaptive performance optimization.

We propose AIAER-Mask, a neuro-symbolic routing framework that separates what must never happen from what should be optimized. The symbolic layer maintains a capability map $C(a, o)$ that declares whether action or replica a may serve operation o. This layer produces a hard mask over invalid actions. The neural layer, implemented as an offline discrete Soft Actor-Critic (SAC) policy [25], ranks candidate actions using latency, queue, load, success-rate, and stress features. The learned policy is permitted to adapt only within the symbolic safety envelope.

The evidence supports a nuanced interpretation: AIAER-Mask provides strict intent correctness through hard symbolic masking under a validated and actively enforced capability map, while strong masked heuristics such as Mask+EWMA and Intent-safe P2C can remain competitive or outperform the learned policy in some latency and regret regimes. The strongest contribution is that hard symbolic masking provides strict intent correctness at very low masking cost, while the learned policy offers a path toward adaptive safety-constrained routing as environments become larger, more dynamic, and more non-stationary.

To assess whether the symbolic intent-guard design is tied to a single application topology, the evaluation includes DeathStarBench

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HotelReservation in addition to SocialNetwork. HotelReservation has a different operation set and service structure, so it tests whether workload-specific capability maps can instantiate the same hard-mask mechanism beyond the SocialNetwork setup. This extension strengthens applicability evidence, but it does not imply universal applicability to all microservice systems.

The paper makes the following contributions:

1) *Safe neuro-symbolic routing formulation:* We formalize intent-correct event routing using a capability map and a hard action mask. The symbolic mask is deterministic and lightweight; it is not an AI model.
2) *Offline learning inside a safe action space:* We combine hard symbolic masking with offline discrete SAC so that the learned router optimizes telemetry-driven decisions only among semantically valid replicas.
3) *Comprehensive evaluation pipeline:* We evaluate ablations, masked versus unmasked variants, legacy baselines, failure injection, dynamic/adversarial scenarios, broker-delay and Local Region-Latency Emulation, stress calibration, hyperparameter sensitivity, operational complexity, and reproducibility across five seeds.
4) *Cross-workload applicability evidence:* We evaluate the framework on two DeathStarBench workloads, SocialNetwork and HotelReservation, showing that workload-specific symbolic masks preserve intent correctness across both workloads while maintaining honest comparisons against strong masked heuristics.
5) *Honest safety-performance interpretation:* We show that masking produces large correctness gains and invalid-action prevention, while most extra overhead in full AIAER-Mask comes from neural inference and state processing rather than from the symbolic mask itself.

The remainder of the paper is organized as follows. Section II reviews related work. Section III defines the system model and safety constraint. Section IV presents the architecture. Section V describes the masked offline policy and data pipeline. Section VI describes the experimental methodology. Section VII presents results. Section VIII discusses operational complexity, applicability, limitations, and future work. Section IX concludes.

II. Related Work

A. Microservice and Event-Driven Routing

Microservice routing is commonly implemented through static tables, round-robin, least-connections, EWMA latency routing, and randomized load balancing such as Power-of-Two-Choices [2]. Heuristic and metaheuristic load-balancing methods remain attractive because they are inexpensive and transparent [7]. In event-driven systems, message brokers such as NATS, Kafka, or RabbitMQ add decoupling between producers and consumers, but they do not automatically solve semantic routing correctness. A broker may deliver an event efficiently while the application-level router still chooses a semantically invalid handler.

B. Reinforcement Learning for Cloud Systems

Reinforcement learning has been studied for cluster scheduling, resource allocation, load balancing, and microservice placement [5], [6], [8], [9]. These systems can adapt to changing load, but online exploration can be unsafe in production. More importantly for this work, many formulations optimize performance metrics over anonymous requests and do not encode semantic operation constraints. An RL router without a hard safety layer may learn that an invalid but fast action is attractive.

C. Offline RL and Safe Deployment

Offline RL learns from logged data without online exploration [3], [4]. Conservative methods such as CQL, IQL, and TD3+BC reduce out-of-distribution action selection [10], [11], [12]. Off-policy evaluation and Fitted Q Evaluation help screen policies before deployment [13], [14]. These techniques improve deployment discipline, but they do not by themselves guarantee semantic admissibility. AIAER-Mask uses offline RL for adaptation, while a separate symbolic guard enforces admissibility.

D. Safe RL, Constrained Decisions, and Action Masking

Safe RL and constrained decision systems distinguish reward optimization from constraint satisfaction. Invalid action masking has been shown to improve learning in structured action spaces by removing illegal choices from the policy distribution [18], [19], [20], [21]. Our setting differs from games or schedulers where invalidity is often syntactic. Here, invalidity is semantic and application-specific: an action is valid only if the service replica is capable of handling the operation intent.

E. Intent-Based and Neuro-Symbolic Systems

Intent-based networking maps operator goals to network behavior [15], [16], [17]. Neuro-symbolic systems combine learned models with symbolic constraints or reasoning layers [22], [23], [24]. AIAER-Mask follows this hybrid design principle, but applies it to event-driven microservice replica selection. The symbolic component is deliberately simple and auditable; the neural component is used for performance adaptation inside the safe region.

F. MLOps and Reproducible AI Systems

Deploying learned policies in systems infrastructure introduces operational concerns beyond model accuracy. MLOps work emphasizes versioned data pipelines, monitored deployment, reproducible environments, and model lifecycle management [26], [27], [28]. These concerns are directly relevant to learned routing: stale telemetry, topology drift, capability-map changes, and policy-update cadence can all affect safety and performance. The experiment pipeline therefore records environment details, seeds, repeated trials, dataset provenance, warm-up filtering, hyperparameter sensitivity, and operational-complexity measurements.

III. System Model and Safety Formulation

We model an event-driven application as a set of operation types $\mathcal{O}$ and candidate routing actions $\mathcal{A}$. An action corresponds to a service replica or broker subject endpoint. The implementation uses a multi-replica capability map, so an operation can have multiple valid replicas:

$$V(o) = \{a \in \mathcal{A} : C(a, o) = 1\}. \tag{1}$$

For example, `compose_post` may be served by `compose_post_r1`, `compose_post_r2`, or `compose_post_r3`. Replicas within the same operation group are semantically valid, but they have different latency, queue sensitivity, jitter, and degradation behavior.

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The capability function is

$$C(a, o) \in \{0, 1\}, \tag{2}$$

where $C(a, o) = 1$ means that action a is semantically admissible for operation o. The intent-correctness property is

$$\forall t : \ C(a_t, o_t) = 1. \tag{3}$$

Equivalently, a policy π is intent-correct if

$$\Pr_{\pi}[\forall t : \ C(a_t, o_t) = 1] = 1. \tag{4}$$

This notation makes the quantifier explicit and avoids ambiguity about whether correctness is checked per decision or only in aggregate.

The router observes a state vector s_t containing operation identity, valid-action mask, recent latency estimates, queue-depth proxies, estimated load, recent success rates, stress level, broker-delay setting when used, and replica metadata. The routing objective is to optimize latency and reliability within the constraint:

$$; \max_{\pi} \ \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right]$$
$$\text{s.t.} \ \Pr_{\pi}[\forall t : \ C(a_t, o_t) = 1] = 1. \tag{5}$$

The reward includes latency, success, timeout, invalid-action penalty for unsafe variants, and regret-aware shaping during training where telemetry permits estimating the best valid replica. Intent correctness is enforced by the mask rather than by reward shaping.

If $V(o) = \emptyset$, the safe behavior is not to guess. The router rejects the request or invokes the configured fallback path and records the fallback. This behavior is evaluated in failure-injection scenarios.

Proposition 1(Intent correctness under hard symbolic masking): Assume that (i) the capability map $C(a, o)$ is correct for the active deployment configuration, (ii) the router applies the hard mask $M(o)$ before every action selection, (iii) the action selector samples or chooses only from the masked action set, and (iv) unknown operations or empty valid-action sets are rejected or sent to fallback rather than guessed. Then every selected non-fallback action a_t satisfies $C(a_t, o_t) = 1$.

Proof sketch: For every invalid action a with $C(a, o_t) = 0$, the hard mask removes the action from the admissible set or assigns it a blocking logit. The stochastic policy therefore assigns zero probability to invalid actions, and a deterministic selector cannot choose them because it operates only over $V(o_t)$. Thus, an invalid non-fallback action can be selected only if the capability map is wrong, the mask is bypassed, or an implementation fault violates the assumptions. The argument is a formal safety argument for the implemented masking rule, not a machine-checked proof of the whole deployment. Full formal verification remains future work.

IV. AIAER-Mask Architecture

A. Separation of Symbolic and Neural Layers

AIAER-Mask has two distinct layers:

- *Symbolic safety layer:* This layer stores the capability map, validates operation-action admissibility, constructs the hard mask, and blocks invalid cross-operation routing. It is deterministic, rule-based, auditable, and lightweight.
- *Neural adaptive layer:* This layer is an offline-trained discrete SAC policy. It receives telemetry-derived state features and selects among the valid replicas that remain after masking.

The symbolic mask is not an AI model. It is a lightweight symbolic safety layer that constrains the action space before the learned policy acts.

Fig. 1 summarizes the proposed neuro-symbolic routing pipeline. The symbolic layer first filters the action space through the capability map, after which the learned policy ranks only admissible replicas using telemetry-derived state features.

B. Hard Masked Policy

Given operation o_t, the symbolic guard constructs a mask $M(o_t)$:

$$M_a(o_t) = \begin{cases} 0, & C(a, o_t) = 1, \\ -\kappa, & C(a, o_t) = 0, \end{cases} \tag{6}$$

where κ is a large constant. For policy logits $z_\theta(s_t, a)$, the masked policy is

$$\pi_\theta(a|s_t, o_t) = \frac{\exp(z_\theta(s_t, a) + M_a(o_t))}{\sum_{a' \in \mathcal{A}} \exp(z_\theta(s_t, a') + M_{a'}(o_t))}. \tag{7}$$

If $C(a, o_t) = 0$, the probability of selecting a is numerically zero. This is the core reason hard masking can guarantee intent correctness when the capability map is correct and actively applied.

C. Training Stabilization

The offline learner uses discrete SAC with behavior-cloning regularization and regret-aware reward shaping. Behavior cloning is applied only to logged behavior actions that are valid under the active capability map. Regret-aware shaping uses telemetry snapshots in the baseline behavior logs to estimate the latency gap between the selected action and the best valid replica. No final-evaluation outputs are used for training.

V. Dataset, Implementation, and Reproducibility

A. Implementation Platform

The implementation uses DeathStarBench SocialNetwork and HotelReservation [1] with Dockerized services and a NATS-based event fabric [29]. SocialNetwork is the primary workload for the full experiment pipeline. Its logical action space contains 18 actions across 6 operation groups, with 3 valid replicas per operation. The SocialNetwork capability map is externalized in a YAML configuration file. Validation reports 6 operations, 18 actions, 18 valid operation-action pairs, 90 invalid pairs, no unmapped operations, and no zero-valid-action operation. HotelReservation uses a separate workload-specific YAML capability map to evaluate whether the symbolic mask can be instantiated for a second DeathStarBench service topology. The validator emits a map hash and machine-readable report, and the evaluated maps include a `schema_version` field so that capability maps can be treated as versioned deployment artifacts.

B. Offline Dataset Collection

The offline dataset is built only from dedicated behavior-policy baseline CSVs, not from final evaluation logs. The behavior policies include Intent-EWMA, H-LRU, H-A$*$, Op-Static, P2C, LEW, and Intent-safe P2C under low, medium, and high stress levels, with five seeds. Additional training perturbations include queue-depth spike, regional latency spike, and burst load under high stress. The data collection report records 210 baseline CSV files and 27,227,991 loaded rows. To keep the monolithic training artifact manageable, the builder writes a deterministic 1,000,000-transition training sample from the available baseline transitions and records the sampling metadata. Logged fields include operation, selected action, latency, success, overhead, timeout, application error, active and true capability validity, invalid selections, fallback indicators, warm-up flags, and telemetry snapshots.

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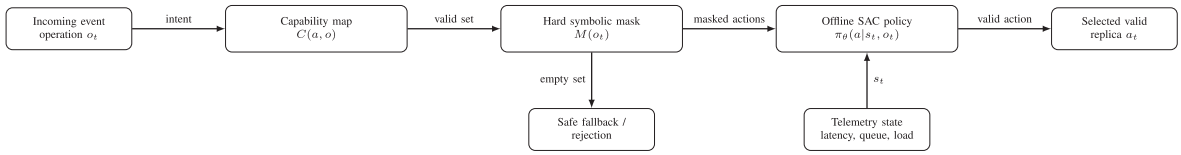


Fig. 1. Neuro-symbolic routing workflow. The symbolic layer determines which actions are admissible. The neural layer ranks only those actions using telemetry state.

TABLE I
ANALYTICAL COMPARISON WITH ROUTING, RL, SAFE-RL, SYMBOLIC, AND NEURO-SYMBOLIC APPROACHES. THE YEAR COLUMN PROVIDES CONTEXTUAL METADATA ONLY.

Work / Family	Year	Routing Type	Learning	Symbolic	Intent-Aware	Dynamic	Safety
Static, EWMA, P2C heuristics [2], [7]	2001–2025	Heuristic replica selection	No	No	No	Partial	No
RL cloud routing and scheduling [5], [6], [9]	2022–2024	Adaptive performance routing	Yes	No	No	Yes	No
Offline RL and OPE [3], [10], [14]	2020–2022	Offline policy learning	Yes	No	No	Partial	No hard semantic guarantee
Intent-based networking [15], [16]	2023–2024	Policy-driven network control	Partial	Policy-level	Yes	Partial	Depends on controller
Invalid action masking [18], [19], [21]	2023–2026	Constrained action selection	Often	Yes	Domain-specific	Partial	Yes, if mask is correct
Neuro-symbolic RL [22]–[24]	2020–2025	Learned policy with symbolic constraints	Yes	Yes	Domain-specific	Yes	Depends on specification
AIAER-Mask	**This work**	**Intent-correct microservice routing**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes, under validated map**

TABLE II
HARDWARE ENVIRONMENT USED FOR THE REPORTED EXPERIMENTS.

Component	Specification
CPU	Intel Core ultra 9 275hx
GPU	NVIDIA GeForce RTX 5080 Laptop GPU
RAM	64 GB
Storage	2TB PCIe Gen 4 SSD
OS	Windows 11

TABLE III
SOFTWARE STACK RECORDED FOR REPRODUCIBILITY.

Software	Version
Python	3.10.11
PyTorch	2.10.0.dev20250921+cu128; CUDA-enabled local development build
Docker Desktop	4.73.1
Docker Engine/Server	29.5.2
NumPy	1.26.4
pandas	2.2.2
matplotlib	3.8.4
NATS	Docker service; image nats:2.10
DeathStarBench	commit 6ecb09706140

C. Software and Hardware

Tables II–IV summarize the execution environment and experiment configuration. Values are taken from the generated reproducibility report and configuration files.

D. Warm-up and Reporting

Each trial records warm-up rows using an `is_warmup` column. Aggregation excludes warm-up rows and generates a warm-up filtering report. Summary files report means, standard deviations, and confidence intervals where applicable. The pipeline also generates LaTeX-ready tables, figure audits, experiment audit reports, and reproducibility reports.

TABLE IV
KEY EXPERIMENTAL CONFIGURATION.

Parameter	Value
Seeds	42, 123, 2024, 2025, 999
Independent trials	5 per configuration
Stress tiers	5k, 9k, 14k requests/s
Warm-up	10 s and 100 warm-up requests
Measurement phase	60 s per trial
Baseline training logs	210 CSV files
Source rows loaded	27,227,991
Training transitions written	1,000,000 sampled from baseline logs
Dataset split	70/15/15 train/validation/test
Action space	18 replica actions
State dimension	168
Ablation training steps	3000
Max ablation samples	300,000
Batch size / hidden units	256 / 256

VI. EXPERIMENTAL METHODOLOGY

A. Research Questions

The evaluation addresses the following questions:

- *RQ1:* Does hard symbolic masking improve intent correctness and invalid-action prevention?
- *RQ2:* Does the mask itself introduce meaningful operational overhead?
- *RQ3:* How does AIAER-Mask compare with strong masked heuristics in stable and dynamic regimes?
- *RQ4:* How robust is the hard-mask design under failure injection and stale or abnormal conditions?
- *RQ5:* What are the reproducibility, hyperparameter, stress-calibration, broker-dependency, and region-latency boundaries?
- *RQ6:* Can the same symbolic masking mechanism be instantiated across more than one DeathStarBench workload?

B. Baselines and Variants

The final comparison includes the full AIAER-Mask model, AIAER-Mask without selected state features, SAC without hard mask, SAC

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with soft intent penalty, Mask+Static, Mask+EWMA, Intent-safe P2C, H-LRU, H-A$*$, P2C, LEW, EWMA without mask, and Static without mask. The no-mask heuristics deliberately select from the full action space and may choose invalid cross-operation replicas. Existing masked heuristics remain constrained by the capability map.

C. Cross-Workload Evaluation

We instantiate AIAER-Mask on two DeathStarBench workloads: SocialNetwork and HotelReservation. Each workload uses its own capability map that defines valid operation-action pairs for that application. The same evaluation principles are applied across workloads: capability-map validation, hard symbolic masking, masked versus no-mask comparison, stable evaluation, safety metrics, and repeated seeded trials where applicable. The goal is not to claim universal applicability, but to test whether the capability-map-based symbolic mask preserves intent correctness across different DeathStarBench microservice structures.

D. Stress, Broker, and Region-Latency Sensitivity

The offered-load tiers are configured as 5k, 9k, and 14k requests/s for low, medium, and high stress. The calibration report validates that these are ordered offered-load tiers and observes latency and p99 spread across tiers. We do not claim these values were derived from a full empirical saturation search. Broker dependency is evaluated through broker-delay emulation at 0, 2, 5, 10, and 20 ms. This is a broker-delay sensitivity study, not a real multi-broker comparison across NATS, Kafka, and RabbitMQ. We also add Local Region-Latency Emulation: services remain local, while deterministic router/environment-layer latency profiles approximate same-region, near-region, far-region, asymmetric, degraded, recovering, stale-telemetry, and bursty region-like conditions. These profiles test timing sensitivity and adaptation under local emulation only; they are not a real multi-region deployment.

E. Dynamic and Safety Scenarios

Dynamic adaptation scenarios include sudden degradation, recovery, queue spike, rotating hotspot, bursty workload, mixed operation pressure, partial failure, oscillating latency, misleading EWMA, simultaneous degradation, and delayed recovery. A separate environment validation report confirms non-stationarity before routing results are interpreted: every scenario is labeled `valid_dynamic`, the oracle best replica changes over time, and best-versus-worst valid-replica latency gaps are measurable.

Safety scenarios include unknown operation label, invalid service capability, service instance failure, stale capability map, burst load, queue-depth spike, and regional/network latency spike. These tests distinguish active capability validity from true capability validity, which is important when capability metadata is stale.

F. Metrics

We report mean latency (ms), p99 latency (ms), throughput (events/s), routing overhead (μs/request), success rate (%), intent correctness (%), invalid action rate (%), invalid action count, fallback count, cumulative regret (ms), optimal replica match (%), adaptation time (ms), and recovery time (ms). Success rate is not the same as intent correctness: a request can complete at the system level while being semantically misrouted.

VII. Results

A. Capability Map and Correctness Validation

The multi-replica capability map validates successfully: 6 operations, 18 actions, 18 valid operation-action pairs, and 90 invalid operation-action pairs. No unmapped operation, zero-valid-action operation, or unassigned service capability is present in the validated map. Under normal evaluation, all hard-masked variants report 100% active capability validity and 0% invalid-action rate.

B. Masked Versus No-Mask Routing

Fig. 2 compares masked and unmasked counterparts. The effect of the symbolic mask is large and direct. Across comparable pairs, masking improves intent correctness by 91.06 percentage points on average and reduces invalid-action rate by 91.06 percentage points on average. The largest improvements are observed for Mask+Static versus Static without mask, AIAER-Mask versus SAC without hard mask, AIAER-Mask versus SAC with soft penalty, and Mask+EWMA versus EWMA without mask.

Fig. 2 shows that hard-masked variants preserve intent correctness across stress levels, while no-mask and soft-penalty variants remain unsafe. This confirms that semantic correctness is enforced by the symbolic mask rather than by reward optimization alone.

Fig. 3 compares latency across ablation variants and stress levels. It shows why performance must be interpreted together with correctness: some methods are fast because they are not constrained to semantically valid replicas.

Table V also shows the safety-performance tradeoff. Some performance-only no-mask methods are fast, but their intent correctness is extremely low. Masked heuristics preserve correctness and remain strong latency baselines. Full AIAER-Mask is not the fastest stable-regime method in the current evaluation. Fig. 3, Table V, and Table VI report complementary views rather than duplicate measurements. Fig. 3 shows the stress-level ablation breakdown across low, medium, and high stress. Table V summarizes the main SocialNetwork stable-regime aggregate after warm-up filtering. Table VI reports the workload-level comparison between SocialNetwork and HotelReservation. Therefore, the rows should be interpreted according to their stated measurement context.

C. Cross-Workload Applicability

Table VI summarizes the cross-workload comparison on DeathStarBench SocialNetwork and HotelReservation. The results reinforce the main safety claim. In both workloads, all hard-masked variants shown in the table preserve 100% intent correctness and 0% invalid routing under workload-specific validated capability maps. Full AIAER-Mask records mean latency of 40.78 ms on SocialNetwork and 40.68 ms on HotelReservation, while maintaining 100% correctness under those maps in both cases. Mask+EWMA and Intent-safe P2C also preserve correctness and remain strong latency baselines. Intent-safe P2C is the fastest listed masked method on average in both workloads.

The no-mask baselines demonstrate why correctness must be reported separately from latency. EWMA without mask records intent correctness of only 17.39% on SocialNetwork and 11.80% on HotelReservation, corresponding to invalid-action rates of 82.61% and 88.20%. Static without mask is even more unsafe on SocialNetwork, with 0.85% correctness and 99.15% invalid actions, and remains unsafe on HotelReservation with 11.65% correctness and 88.35% invalid actions. These results show that the symbolic masking mechanism can

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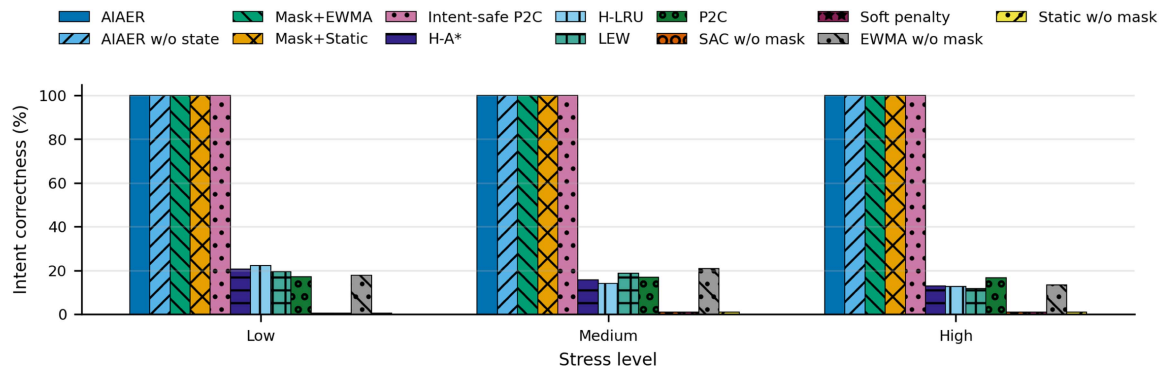


Fig. 2. Intent correctness across ablation variants and stress levels. Hard-masked variants preserve 100% correctness under validated capability maps, while no-mask and soft-penalty variants remain unsafe.

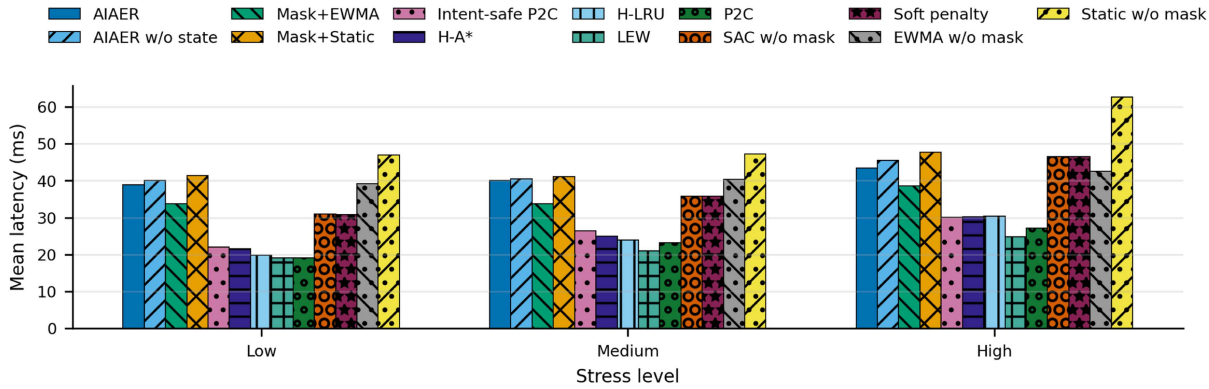


Fig. 3. Mean latency across ablation variants and stress levels. Results should be interpreted jointly with intent correctness and invalid-action rate.

TABLE V

SELECTED STABLE-REGIME AGGREGATE RESULTS FOR THE MAIN SOCIALNETWORK EVALUATION AFTER WARM-UP FILTERING. LOWER LATENCY IS BETTER; HIGHER CORRECTNESS IS BETTER. FIG. 3 REPORTS STRESS-LEVEL ABLATION BREAKDOWNS, WHEREAS THIS TABLE REPORTS AGGREGATE STABLE-REGIME ROWS.

Variant	Mean Latency (ms)	p99 Latency (ms)	Throughput (events/s)	Intent Correctness (%)	Invalid Rate (%)
Static without mask	52.31	59.90	1243.60	0.85	99.15
EWMA without mask	40.72	50.09	1569.82	17.39	82.61
SAC without hard mask	37.75	49.19	1714.32	0.84	99.16
SAC + soft penalty	37.74	49.16	1714.02	0.84	99.16
Intent-safe P2C	26.21	37.91	2473.77	100.00	0.00
Mask + EWMA	35.36	64.49	1817.51	100.00	0.00
Mask + Static	43.40	68.61	1477.95	100.00	0.00
Full AIAER-Mask	40.78	64.77	1543.39	100.00	0.00

TABLE VI

CROSS-WORKLOAD COMPARISON ON DEATHSTARBENCH SOCIALNETWORK AND HOTELRESERVATION. HARD-MASKED VARIANTS PRESERVE 100% INTENT CORRECTNESS AND 0% INVALID ACTIONS UNDER WORKLOAD-SPECIFIC VALIDATED CAPABILITY MAPS, WHILE NO-MASK BASELINES VIOLATE SEMANTIC ROUTING CONSTRAINTS.

Variant	Social Lat. (ms)	Hotel Lat. (ms)	Social Corr. (%)	Hotel Corr. (%)	Social Invalid (%)	Hotel Invalid (%)
Full AIAER-Mask	40.78	40.68	100.00	100.00	0.00	0.00
Mask + EWMA	35.36	36.31	100.00	100.00	0.00	0.00
Mask + Static	43.40	40.77	100.00	100.00	0.00	0.00
Intent-safe P2C	26.21	30.18	100.00	100.00	0.00	0.00
EWMA without mask	40.72	41.88	17.39	11.80	82.61	88.20
Static without mask	52.31	50.42	0.85	11.65	99.15	88.35

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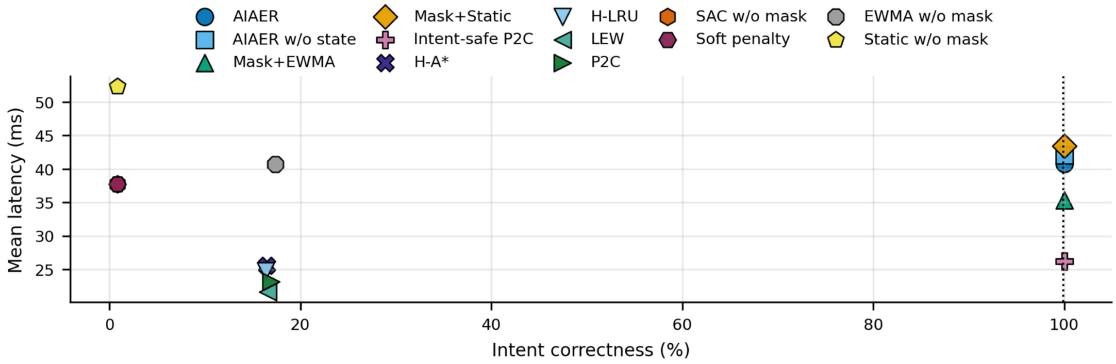


Fig. 4. Safety-performance tradeoff across routing variants. The no-mask methods may be fast but are unsafe; masked methods enforce intent correctness. The 99.9% line is used only as a visual reference for near-strict intent safety and is not an operational threshold selected by the router.

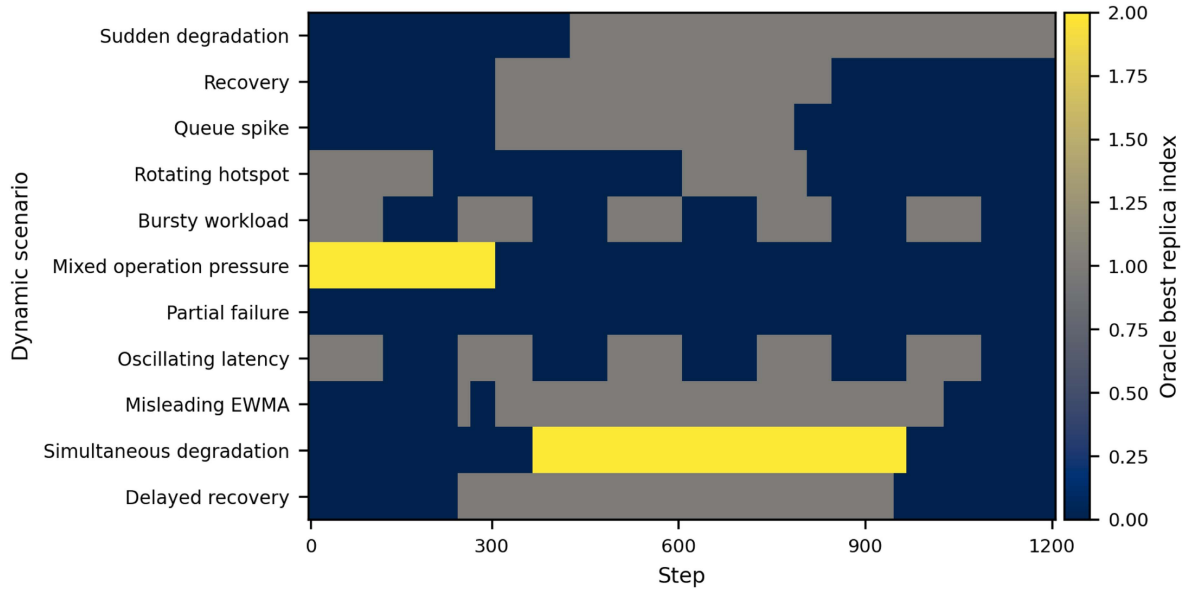


Fig. 5. Oracle best valid replica over time in validated dynamic scenarios. The oracle is used only for evaluation and not for training or routing.

be instantiated across two DeathStarBench workloads, but they do not establish universal applicability to all microservice systems.

D. Safety-Performance Tradeoff Interpretation

The central empirical lesson is that correctness and performance must be interpreted jointly. The masked versus no-mask analysis indicates that simple masked baselines incur an average overhead change of 1.548 μs/request, while the AIAER/SAC hard-mask comparisons show an average overhead change of 335.190 μs/request. This separates the cost of the symbolic intent-guard layer from the cost of the neural adaptive layer.

The correctness guarantee comes primarily from symbolic masking: invalid actions are removed before either a heuristic or learned policy selects a route. The mask is rule-based, deterministic, and capability-map constrained. By contrast, the learned SAC layer contributes adaptive ranking within the valid action set, but also introduces inference, feature-construction, telemetry, and state-processing overhead. This explains why masked heuristics remain extremely competitive in the current results. It also clarifies the expected deployment envelope: as the environment becomes larger, more delayed, more non-stationary, or more multi-region, the value of adaptive learning may increase, but the current single-region setup remains structured enough for strong telemetry heuristics to perform well.

Fig. 4 summarizes the safety-performance tradeoff. The safe variants cluster at 100% intent correctness, whereas performance-only baselines can appear competitive in latency while violating semantic constraints.

E. Dynamic Adaptation

The dynamic environment validation confirms that the tested scenarios are genuinely non-stationary: all scenarios are marked `valid_dynamic`, oracle best-replica paths vary over time, and best-versus-worst valid-replica latency gaps range from approximately 31 ms to 231 ms on average depending on the scenario. Fig. 5 shows the oracle best replica changing over time.

Fig. 5 illustrates that the best valid replica changes over time in the dynamic scenarios. This validates the need to evaluate adaptive routing behavior rather than relying only on static-regime averages.

The dynamic results are important because they prevent overclaiming. In the current setup, Mask+EWMA is the strongest intent-safe dynamic baseline by mean latency, cumulative regret, and optimal-replica

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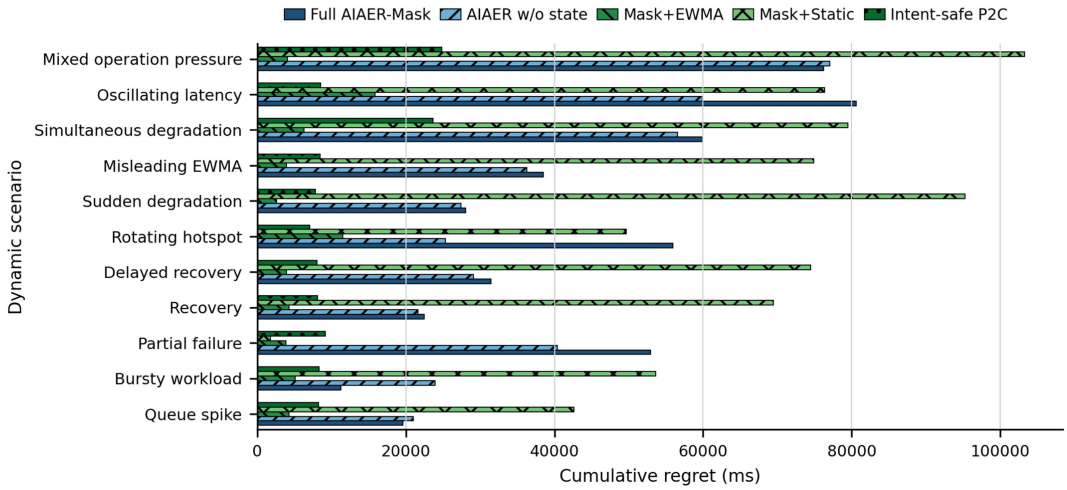


Fig. 6. Cumulative regret for AIAER-family and intent-safe methods in dynamic scenarios. Results should be interpreted jointly with intent correctness and invalid-action rate.

TABLE VII
SELECTED DYNAMIC-REGIME RESULTS. MASK+EWMA REMAINS A STRONG INTENT-SAFE BASELINE IN THE CURRENT DYNAMIC SCENARIOS.

Variant	Mean Latency (ms)	Cumulative Regret (ms)	Optimal Match (%)	Intent Correctness (%)
Mask + EWMA	30.72	5967.10	68.97	100.00
Intent-safe P2C	34.94	11114.41	43.99	100.00
Mask + Static	80.36	65532.98	53.43	100.00
Full AIAER-Mask	88.04	74744.76	44.08	100.00

TABLE VIII
HYPERPARAMETER SENSITIVITY SCOPE. THE SELECTED VALUES ARE DEFAULTS USED FOR THE MAIN CONFIGURED RUN; THE SWEEP CHARACTERIZES SENSITIVITY RATHER THAN FINAL-TEST TUNING.

Hyperparameter	Tested values	Default	Purpose
Learning rate	1e-4, 3e-4, 1e-3	3e-4	Step size
Batch size	128, 256, 512	256	Update stability
Hidden units	128, 256, 512	256	Capacity
γ	0.95, 0.99	0.99	Horizon
Entropy α	0.05, 0.10, 0.20	0.10	Exploration entropy
BC coefficient	0.05, 0.10, 0.20	0.10	Behavior regularization
Regret coefficient	0.30, 0.50, 1.00	0.50	Regret shaping

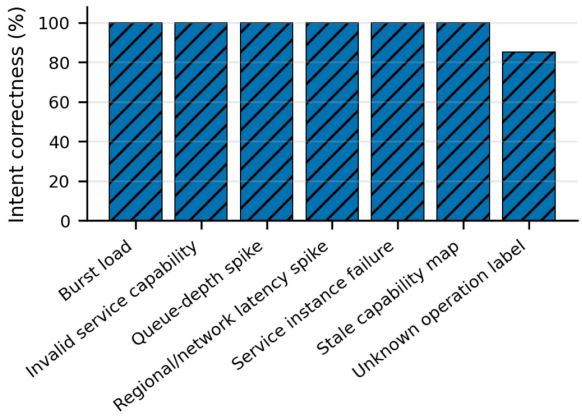


Fig. 7. Intent correctness under failure injection. Known-operation failure scenarios preserve 100% correctness under the active capability map; unknown operations are handled through controlled fallback/rejection.

match. AIAER-Mask preserves correctness, but it does not minimize dynamic regret in every scenario. This suggests that the present environment, although non-stationary, is still simple enough for strong telemetry heuristics to remain effective. The generated drift-robustness artifact also reorganizes the dynamic and failure-injection outputs around governance and drift cases such as stale maps, service failure, queue spikes, rotating hotspots, delayed recovery, misleading EWMA, and simultaneous degradation. These are controlled stress cases rather than production topology drift. Larger action spaces, delayed telemetry, multi-region dynamics, and more complex topology drift may create conditions where learned policies become more valuable.

Fig. 6 reports cumulative regret in dynamic scenarios. It should be read together with Table VII: masked heuristics remain strong in these controlled dynamics, while learned adaptation becomes clearer under harder timing conditions such as Local Region-Latency Emulation.

F. Failure Injection and Safety Robustness

Failure injection confirms that hard-masked routing prevents invalid actions under the active capability map. Across burst load, invalid service capability, queue-depth spike, regional/network latency spike, service instance failure, and stale capability map scenarios, invalid actions selected remain zero and intent correctness remains 100%. The unknown-operation case triggers controlled fallback behavior, yielding a recorded fallback success rate of 100% for fallback requests and an overall success/intent-correctness rate of approximately 84.95% because unknown labels are intentionally not treated as normal valid operations.

Fig. 7 shows that known-operation failure scenarios preserve correctness under the active capability map. Unknown operations are intentionally handled through fallback/rejection rather than guessed routing.

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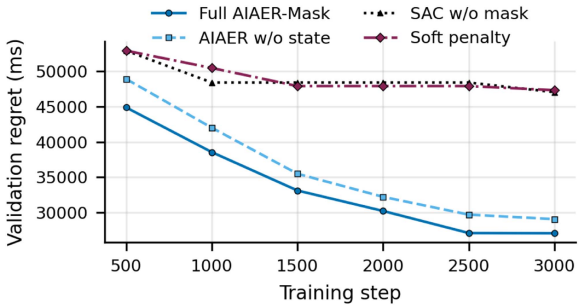


Fig. 8. Validation regret across training steps. Convergence is assessed using held-out validation diagnostics, entropy stabilization, and optimal-valid-replica match rather than reward alone; safety is enforced separately by the hard symbolic mask under a validated capability map.

G. Ablation, Hyperparameters, and Training Convergence

The ablation study isolates the role of hard masking, soft penalties, static masked routing, EWMA masked routing, and reduced state features. The key result is that soft penalties and no-mask SAC do not guarantee safety. Training is evaluated using held-out diagnostics rather than a reward-only curve.

Fig. 8 presents validation regret rather than raw reward. This supports the training-stability discussion, while safety remains enforced separately by Proposition 1 and the hard symbolic mask under the validated capability map.

The training diagnostics are optimization diagnostics rather than safety evidence. Because the reward combines latency, regret, timeout, and regularization terms, average reward can remain negative or visually stable even when validation regret and optimal-match behavior improve. We therefore assess training stability using validation regret, policy entropy, optimal-valid-replica match, critic/Q-loss stability, and held-out routing metrics. Safety is enforced separately by Proposition 1 and the hard symbolic mask under a validated capability map. Final test metrics are not used for hyperparameter tuning.

The hyperparameter study varies learning rate, batch size, hidden units, discount factor, entropy temperature, behavior-cloning coefficient, and regret coefficient. Each configuration records training outcomes and routing metrics. The final pipeline uses a deterministic 70/15/15 train/validation/test split: validation diagnostics support sensitivity analysis, while final test measurements are not used for tuning. The training audit records source-log, configuration, dataset, and capability-map hashes; duplicate-curve checks; entropy; validation regret; optimal-valid-replica match; and invalid-action diagnostics where available. Thus, convergence is interpreted through validation diagnostics and routing metrics rather than reward alone.

H. Generalization, Broker, and Region-Latency Sensitivity

Generalization is evaluated through workload-distribution shifts, cross-workload evaluation, and Local Region-Latency Emulation. The distribution artifact covers uniform, Zipfian, read-heavy, write-heavy, bursty, and mixed operation mixes. Cross-workload evaluation instantiates the symbolic mask on DeathStarBench SocialNetwork and HotelReservation. Local Region-Latency Emulation keeps services local while injecting region-like timing profiles in the router/environment layer. Tables IX and X report bounded local timing and broker-delay emulation; neither is a real multi-region or multi-broker deployment.

Table IX highlights the setting in which the learned policy is most useful. Under Local Region-Latency Emulation, Full AIAER-Mask achieves the lowest cumulative regret among intent-safe variants (63.95 ms) and the highest optimal-valid-replica match (98.29%), while preserving 100% correctness and 0% invalid actions. Compared with Mask+EWMA, this corresponds to approximately 93.1% lower regret and a 4.07 percentage-point higher optimal match. EWMA without masking has lower raw latency, but it is not intent-safe, with only 16.12% correctness and 83.88% invalid actions. This supports adaptive value under harder local timing conditions, not real multi-region readiness.

TABLE IX
LOCAL REGION-LATENCY EMULATION SUMMARY. SERVICES REMAIN LOCAL AND REGION-LIKE TIMING IS INJECTED IN THE ROUTER/ENVIRONMENT LAYER; THIS IS BOUNDED LOCAL EMULATION, NOT REAL MULTI-REGION DEPLOYMENT.

Variant	Mean (ms)	Correct. (%)	Invalid (%)	Regret (ms)	Opt. Match (%)
Full AIAER-Mask	**33.45**	100.00	0.00	**63.95**	**98.29**
Mask+EWMA	34.26	100.00	0.00	930.48	94.22
Mask+Static	35.22	100.00	0.00	1797.60	93.35
Intent-safe P2C	40.49	100.00	0.00	6492.47	65.82
EWMA no mask	28.67	16.12	83.88	805.22	15.33

TABLE X
BROKER-DELAY EMULATION FOR FULL AIAER-MASK. THIS IS DELAY EMULATION ONLY, NOT A REAL NATS VERSUS KAFKA VERSUS RABBITMQ COMPARISON.

Delay (ms)	Mean (ms)	p99 (ms)	Thr. (events/s)	Over. (μs/req.)	Corr. (%)
0	38.94	64.79	1611.80	822.53	100.00
2	38.49	68.80	1628.89	820.20	100.00
5	39.30	68.47	1595.86	817.62	100.00
10	39.18	68.16	1601.60	810.58	100.00
20	50.73	79.63	1245.78	832.69	100.00

Table X reports broker-delay emulation for Full AIAER-Mask. Small emulated delays up to 10 ms are partly absorbed by queueing behavior and measurement variance in this setup, while the 20 ms setting shows clearer latency increase and throughput degradation. Intent correctness remains unchanged across all delay settings because broker delay affects timing, not semantic admissibility under the active capability map.

I. Key Takeaways

Three takeaways follow from the evidence. First, hard symbolic masking converts intent correctness from a reward preference into a routing invariant under the validated capability map. Second, unsafe no-mask routing can appear fast because it may choose semantically invalid replicas. Third, the learned AIAER-Mask layer is most valuable under harder timing conditions: stable settings keep Mask+EWMA, Mask+Static, and Intent-safe P2C strong, while Local Region-Latency Emulation shows clearer adaptive value in regret and optimal-valid-replica match.

The artifact also includes a safety-assurance summary that links each mechanism to its evidence and remaining assumption, including hard masking, fallback handling, map validation, failure injection, and delay emulation.

VIII. Operational Complexity and Discussion

A. Operational Complexity

The operational report records a full AIAER-Mask checkpoint size of approximately 914 KB, training time of 66.98 seconds for the selected run, mean inference overhead of 367.83 μs/request after

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warm-up filtering, p50 overhead of 21.50 μs/request, p95 overhead of 1034.50 μs/request, and p99 overhead of 1287.60 μs/request. A local operational-update microbenchmark over 30 repetitions separates symbolic update costs from learned-policy costs: YAML parsing averages 9.11 ms, capability-map loading averages 7.97 ms, capability validation averages 0.018 ms, mask rebuild averages 0.0026 ms, fallback-table rebuild averages 0.00085 ms, checkpoint loading averages 3.11 ms, policy initialization plus warm-up averages 14.66 ms, single-request inference after warm-up averages 0.230 ms, batch inference for 128 requests averages 0.117 ms, and in-memory policy reference replacement averages 0.0032 ms. The artifact does not implement a production deployment-controller swap or separate fallback-table hot reload; those remain deployment-engineering tasks.

This evidence supports a precise interpretation: the symbolic mask is lightweight, but full learned routing is not free. The extra cost comes mainly from neural inference, telemetry/state construction, and feature processing. A production deployment would need model caching, batched inference or optimized inference, policy update governance, and monitoring for topology or workload drift.

B. Why Use AIAER-Mask?

The strongest reason to use AIAER-Mask is not broad speed dominance. The reason is safety-constrained learning. Pure performance-only routing can be fast but unsafe. Soft penalties reduce but do not eliminate invalid actions. Hard symbolic masking eliminates invalid actions for known operations under the active capability map. The learned layer then offers an adaptive routing mechanism within the safe set.

The present results also show that masked heuristics are serious baselines. In stable environments and in the current dynamic scenarios, Mask+EWMA and Intent-safe P2C can match or outperform AIAER-Mask on latency or regret while preserving correctness. This does not weaken the symbolic masking contribution; rather, it clarifies that the mask is broadly useful across both heuristic and learned routers.

C. Practical Implications

For practitioners, the safest near-term deployment pattern is layered. A symbolic capability guard should be placed in front of any performance optimizer, including simple heuristics, because it prevents entire classes of semantic routing failures. If a system is relatively stable and telemetry is fresh, masked heuristics may provide an attractive low-overhead operating point. The full learned AIAER-Mask policy is most appropriate when the routing environment is sufficiently dynamic to justify the additional inference and feature-processing cost, and when the operator can support model monitoring, offline retraining, and capability-map governance.

D. Applicability

AIAER-Mask applies best to systems where operations have clear semantic routing constraints and where invalid routing has real correctness, privacy, consistency, or compliance cost. The cross-workload extension strengthens evidence that AIAER-Mask is not specific to one DeathStarBench topology, because SocialNetwork and HotelReservation use different operation sets and workload-specific capability maps. However, broader production traces and multi-region deployments remain future work. The framework is less compelling for homogeneous stateless services where every replica is truly interchangeable, and it requires a maintained capability map, telemetry collection, and an offline training pipeline for the learned component.

E. Capability-Map Evolution

The current implementation treats the validated capability map as the source of routing safety during an experiment. If an operation or action is not present in that map, the router does not infer a new capability at runtime. It rejects the event or sends it to the configured fallback path and records the context for analysis. This conservative behavior is deliberate: allowing a learned policy to promote an unknown action directly into the active safety map would weaken the intent-correctness guarantee.

Capability-map validation checks that every operation has at least one valid action, every valid action exists in the action space, every action belongs to a declared service capability group, no duplicate or unknown actions appear, no service action is left unassigned, and valid/invalid pair counts are nonempty. The generated validation artifact records hashes for the SocialNetwork and HotelReservation maps, supporting CI-style version tracking. In deployment, CI should reject zero-valid actions, unknown actions, duplicate mappings, missing schema fields, or owner mismatches. Capability-map changes should require service-owner approval, CI validation, shadow/canary testing, runtime monitoring, and rollback. AI-assisted discovery may cluster repeated unknown-operation events and propose candidate updates, but candidates must remain inactive until schema checks, policy rules, service-owner review, shadow evaluation, and rollback planning succeed. Thus, learning may help maintain the map while the active hard mask remains deterministic and auditable.

F. Limitations and Future Work

The present study provides bounded evidence rather than universal deployment claims. It evaluates two DeathStarBench workloads, SocialNetwork and HotelReservation, which improves workload diversity but remains benchmark-based rather than production-trace validation. All experiments are single-region; Local Region-Latency Emulation injects controlled region-like timing asymmetry and therefore strengthens timing stress coverage, but it does not replace real multi-region deployment studies. Broker sensitivity is likewise delay emulation, not a direct NATS versus Kafka versus RabbitMQ deployment comparison.

The dynamic and failure-injection scenarios are validated and nonstationary, but they remain controlled experiments rather than production topology drift at scale. The current environment is also relatively structured, and strong masked telemetry heuristics remain highly competitive. Thus, the learned router should be viewed as a safety-constrained adaptive framework whose benefits are expected to become more pronounced in harder delayed-telemetry, multi-region, larger-scale, or stronger-drift settings. Future work should therefore examine additional workloads, production-derived traces, real multi-region deployments, real broker ecosystems, larger action spaces, optimized inference, and more efficient telemetry pipelines.

Applicability depends on explicit semantic routing constraints and maintainable capability maps. Although the map is validated and versioned, correctness of the real-world specification remains a governance responsibility. AI-assisted map evolution is promising, but candidate updates should remain outside the active routing path until validated through schema checks, policy checks, service-owner review, and safe rollout. Taken together, we claim intent correctness under the active validated capability map and tested scenarios, but not universal latency superiority, real multi-broker generalization, or real multi-region production readiness.

IX. Conclusion

This paper presented AIAER-Mask, a safe neuro-symbolic routing framework for event-driven microservices. The evaluation shows that

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hard symbolic masking preserves 100% intent correctness and 0% invalid actions for masked variants under validated and actively enforced capability maps. The cross-workload evaluation on DeathStarBench SocialNetwork and HotelReservation, together with Local Region-Latency Emulation, provides bounded evidence that the symbolic intent-guard mechanism can be instantiated across different benchmark service topologies and region-like timing conditions. The symbolic mask itself is lightweight and computationally cheap, especially for heuristic routers. Full learned routing adds additional overhead due to neural inference and telemetry/state processing. Strong masked heuristics remain competitive and sometimes outperform the learned policy in latency or dynamic regret.

The appropriate conclusion is therefore nuanced: AIAER-Mask is not the fastest router in every regime. It is a safety-constrained routing framework that combines deterministic symbolic correctness guarantees with adaptive AI routing inside the safe action space. In future deployments, learning may also support capability-map maintenance by proposing candidate updates for repeated unknown actions, but such candidates must remain outside the active safety map until validated. This framing is more mature and more useful for production event-driven systems where semantic correctness is a first-order requirement.

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